

- 6 (a) packet/quantum of energy.....M1
 energy = hf A1 [2]
- (b) e.g. threshold frequency outlined
 max. k.e. independent of intensity
 max. k.e. dependent on frequency (n.b. NOT proportional)
 photoelectric current depends on intensity
 instantaneous emission (1 each, max 3)..... B3 [3]
- (c) (i) photons have same energy so $E_{\max}$ unchanged
 intensity OR number of photons per unit time is halved,
 so $\frac{1}{2}n$ OR n reduced..... B1
 (allow 1 mark for statement that $E_{\max}$ unchanged and n reduced)
- (ii) photons have higher energy so $E_{\max}$ increases..... B1
 but fewer photons per unit time so n decreases B1 [4]
 (allow 1 mark for statement that $E_{\max}$ increases and n reduced)
 (allow any argument based on increased efficiency)